LAB 8: Server Configuration: DHCP, DNS and Web Server in Cisco Packet Tracer

1. Objectives

- Configure a server to provide DHCP, DNS, and Web (HTTP) services in Cisco Packet Tracer.
- Understand how DHCP dynamically assigns IP addresses and how DNS resolves domain names within a network.
- Verify proper functioning of all server services by accessing the web server from client devices using a domain name.

2. Theory

2.1 Server Configuration

A server is a network device or software system that provides services to client machines on the network. Servers are typically assigned a static IP address so that clients can always reach them at a fixed, known address. In this lab, a single server is configured to simultaneously run three services: DHCP, DNS, and HTTP. This multi-service setup demonstrates how centralized server-based services are deployed in real networks to simplify administration and reduce the number of physical devices required.

2.2 DHCP (Dynamic Host Configuration Protocol)

DHCP is an application-layer protocol that automates the assignment of IP configuration to client devices on a network. Without DHCP, each device would require manual configuration of its IP address, subnet mask, default gateway, and DNS server address, which is impractical in large or frequently changing networks.

The DHCP process uses a four-step exchange called DORA:

- Discover: The client broadcasts a DHCPDISCOVER message to find an available DHCP server.
- Offer: The server responds with a DHCPOFFER containing an available IP address and configuration.
- Request: The client broadcasts a DHCPREQUEST to accept the offered parameters.
- Acknowledge: The server sends a DHCPACK to confirm the lease and complete the configuration.

The server maintains an address pool and assigns leases for a configured time period. Once a lease expires, the address is returned to the pool for reassignment.

2.3 DNS (Domain Name System)

DNS is a hierarchical, distributed naming system that translates human-readable domain names such as www.hcoe.com into their corresponding IP addresses. Without DNS, users would need to remember the IP address of every resource they want to access. DNS operates at the Application Layer and uses both UDP and TCP on port 53.

The most basic DNS record is the A record, which maps a domain name directly to an IPv4 address. When a client types a domain name in a browser or uses it in a ping command, its operating system sends a DNS query to the configured DNS server. The server looks up its records, finds the matching IP address, and returns it to the client so the connection can proceed.

2.4 Web Server (HTTP Service)

A web server delivers web content to clients using the Hypertext Transfer Protocol (HTTP), which operates on TCP port 80. The server hosts HTML files and other web resources and responds to GET requests sent by browsers. In Cisco Packet Tracer, enabling the HTTP service on a server device allows client PCs to load a web page through the built-in browser application, simulating real-world web hosting and access.

2.5 Integration of DHCP, DNS, and Web Server

These three services operate as a chain. DHCP first assigns the client an IP address and tells it the address of the DNS server. The client then queries DNS to resolve the web server domain name into an IP address. Finally, the client uses HTTP to request and receive the web page. Each service depends on the previous one, which means a misconfiguration at any step will break the entire process.

3. Network Topology

The topology consists of Server0, which holds a static IP address of 192.168.1.2, connected through Switch0 to two client PCs named PC0 and PC1. Both PCs are set to receive their network configuration automatically via DHCP.

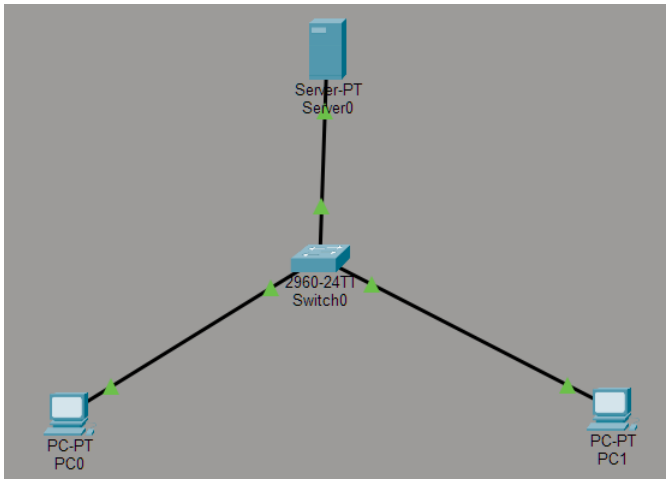

Figure 1: Network Topology — Server0 providing DHCP, DNS, and HTTP services to PC0 and PC1 through Switch0

4. IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server0	192.168.1.2	255.255.255.0	192.168.1.1
PC0	Auto (DHCP)	255.255.255.0	192.168.1.1
PC1	Auto (DHCP)	255.255.255.0	192.168.1.1

5. Configuration Procedure

5.1 Step 1: Assign Static IP to Server

1. Click on Server0 and go to Desktop then IP Configuration.
2. Enter IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, DNS Server: 192.168.1.2.

5.2 Step 2 and 3: Configure DHCP Service

3. Go to Server then Services then DHCP and switch the service ON.
4. Create a new pool using the following parameters and click Add:

Field	Value
Pool Name	LAB_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Verification: On each PC go to Desktop then IP Configuration and select DHCP. The PC should receive an IP address automatically starting from 192.168.1.10.

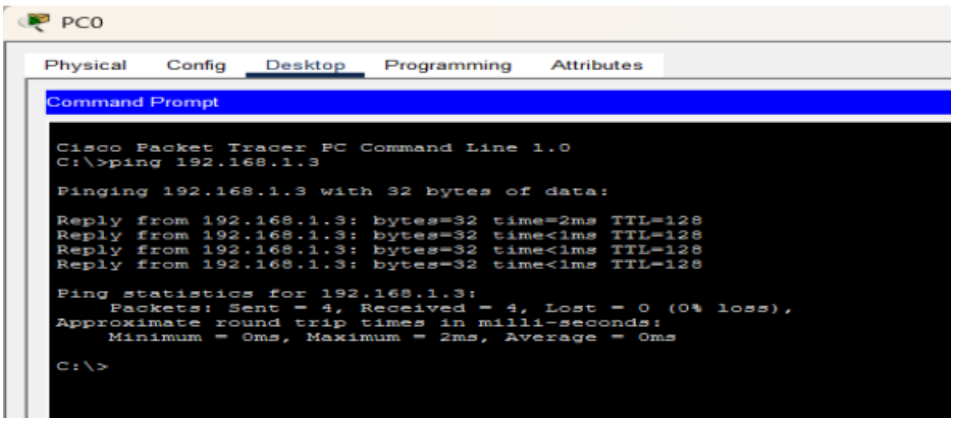

Figure 2: PC0 successfully receiving IP configuration automatically via DHCP from the server

5.3 Step 4 and 5: Configure DNS Service

5. Go to Server then Services then DNS and switch it ON.

6. Add an A record with Name: www.hcoe.com, Type: A Record, Address: 192.168.1.2, then click Add.

Verification: Open Command Prompt on each PC and ping www.hcoe.com. The name should resolve to 192.168.1.2 with all four replies received successfully.

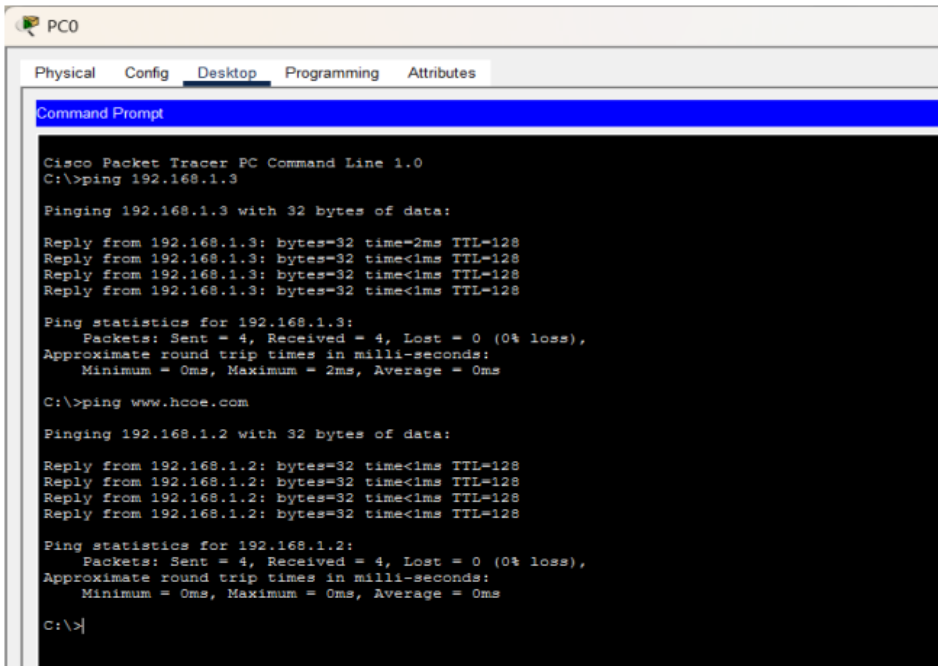

Figure 3: DNS resolution working — pinging www.hcoe.com resolves to 192.168.1.2 from PC0

5.4 Step 6: Configure Web Server (HTTP)

7. Go to Server then Services then HTTP and switch it ON.

8. Optionally edit the index.html file to change the default web page content.

Verification: On any client PC open the Web Browser from Desktop and type http://www.hcoe.com in the address bar. The web page should load successfully.

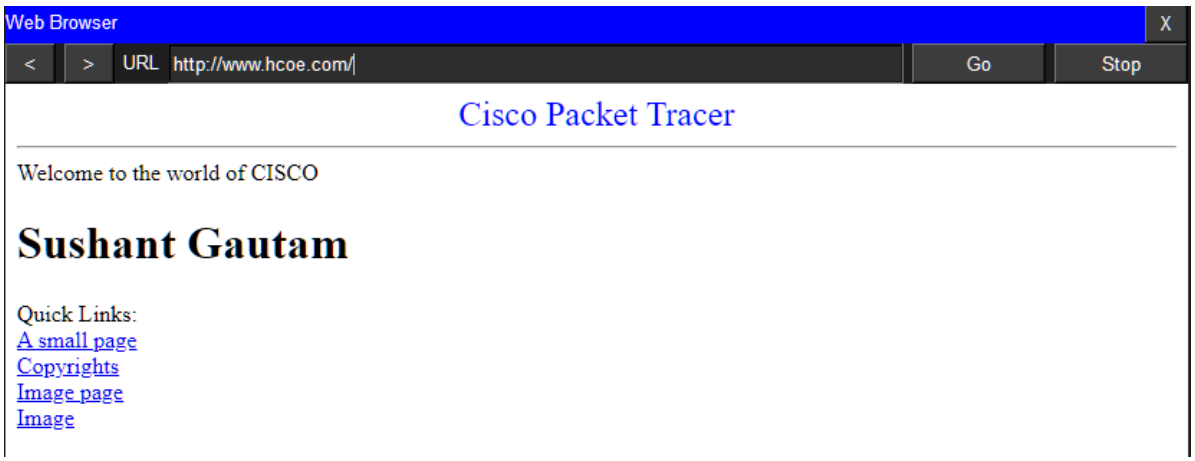

Figure 4: Web page accessed successfully at http://www.hcoe.com through the Packet Tracer browser on PC1

6. Result

All three services were successfully configured and verified. The DHCP server automatically distributed valid IP configurations to both client PCs. DNS correctly resolved the domain name www.hcoe.com to the server IP address 192.168.1.2. The HTTP service delivered the hosted web page to client browsers without any errors, confirming that all services are functioning correctly together.

7. Discussion and Conclusion

This lab demonstrated the practical deployment of three essential network services and their interdependency. DHCP removed the need for manual IP configuration on client devices, DNS made network resources accessible by name rather than by IP address, and HTTP enabled web content delivery across the network.

An important observation from this lab is the service dependency chain. If DHCP failed to provide the correct DNS server address to the client, DNS resolution would fail and the web page would be unreachable. If DNS was not configured, the browser could not translate the domain name to an IP and the HTTP request would never reach the server. This confirms that in real networks, each service must be correctly configured before the next one can work.

The objectives of the lab were fully achieved. All three services were successfully configured, verified through ping and browser tests, and their integration was demonstrated in a simulated network environment using Cisco Packet Tracer.